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World Models & World Action Models

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01

From VLA to World Models

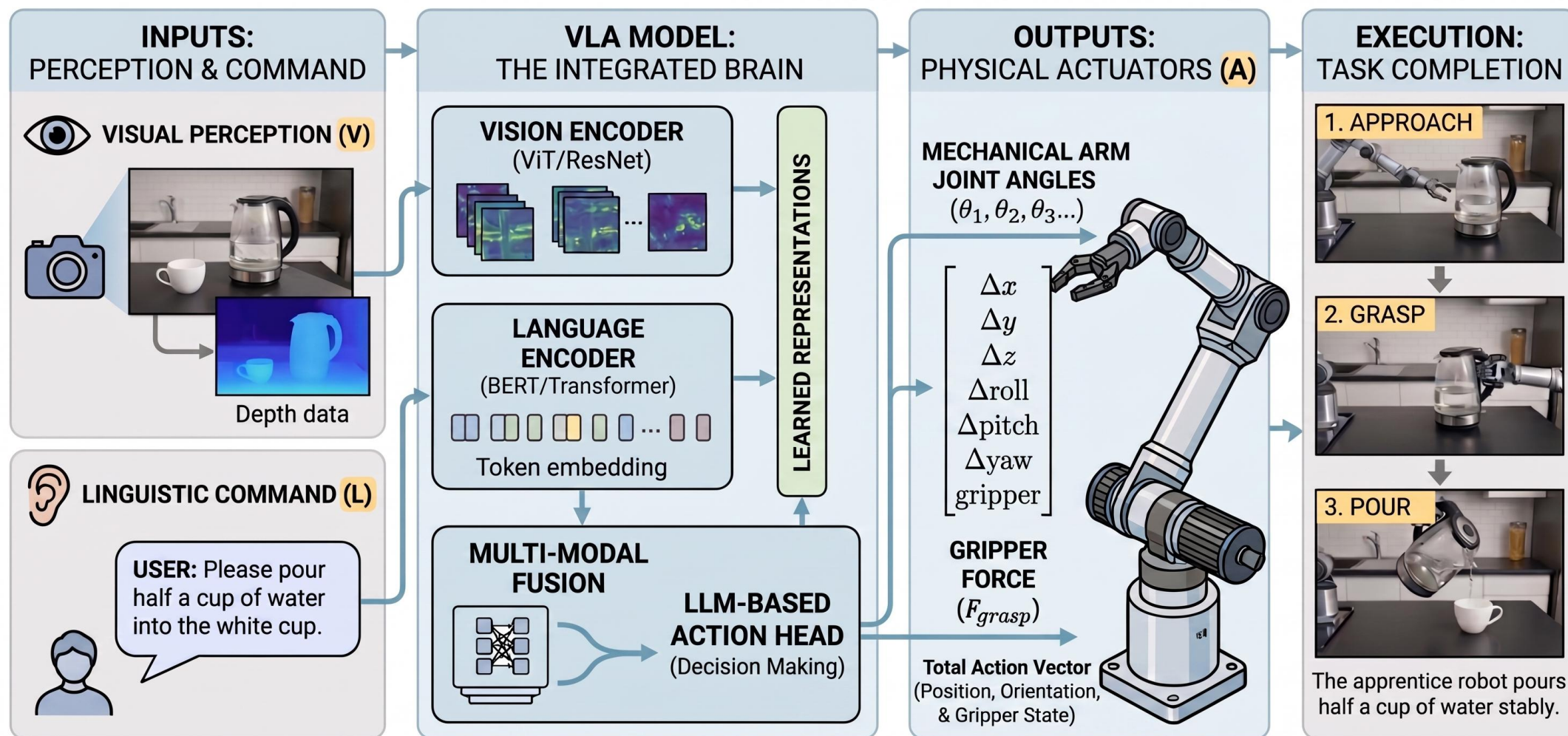
Recap: Vision-Language-Action



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□ **VLA:** Observation o + Linguistic Instructions $l \rightarrow [a_1, a_2, \dots, a_T]$

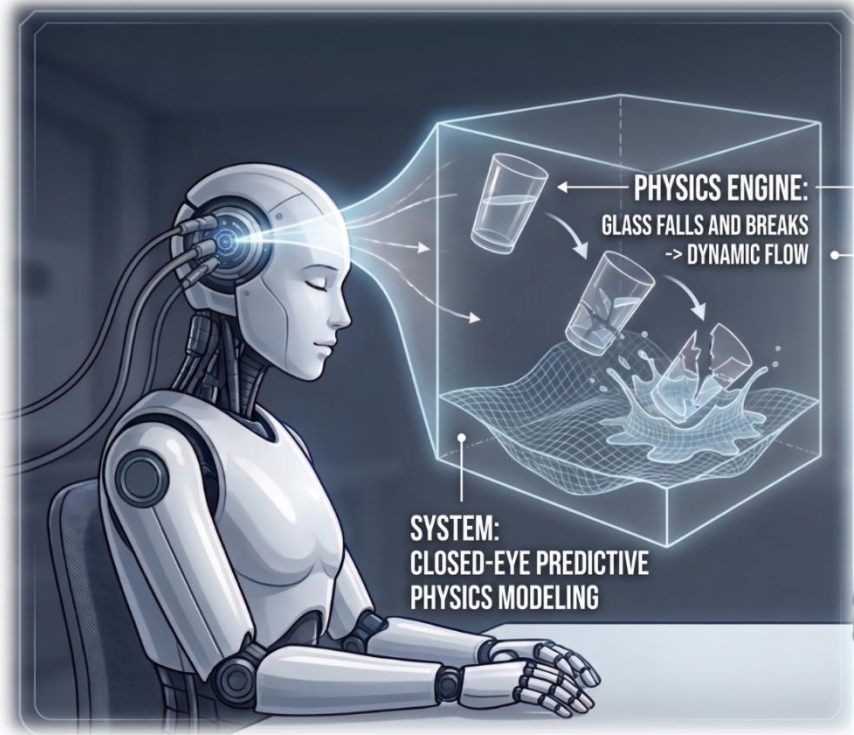




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What is World Models?



❑ **State-transition model:** current state & action $\rightarrow$ next state or a sequence of future states

❑ **Two directions (for robotic policies):**

✦ **Tighter Coupling:** predictive modeling + action generation. **WAM**

✦ **Simulators:** Using world models as simulators for validation, post-training, and RL.

❑ **Equation:** $\mathcal{L}_{WM} = \mathbb{E}_{(o,a,o') \sim \mathcal{D}} [-\log p(o'|o, a)]$

❑ **Three core capabilities of actionable world model**

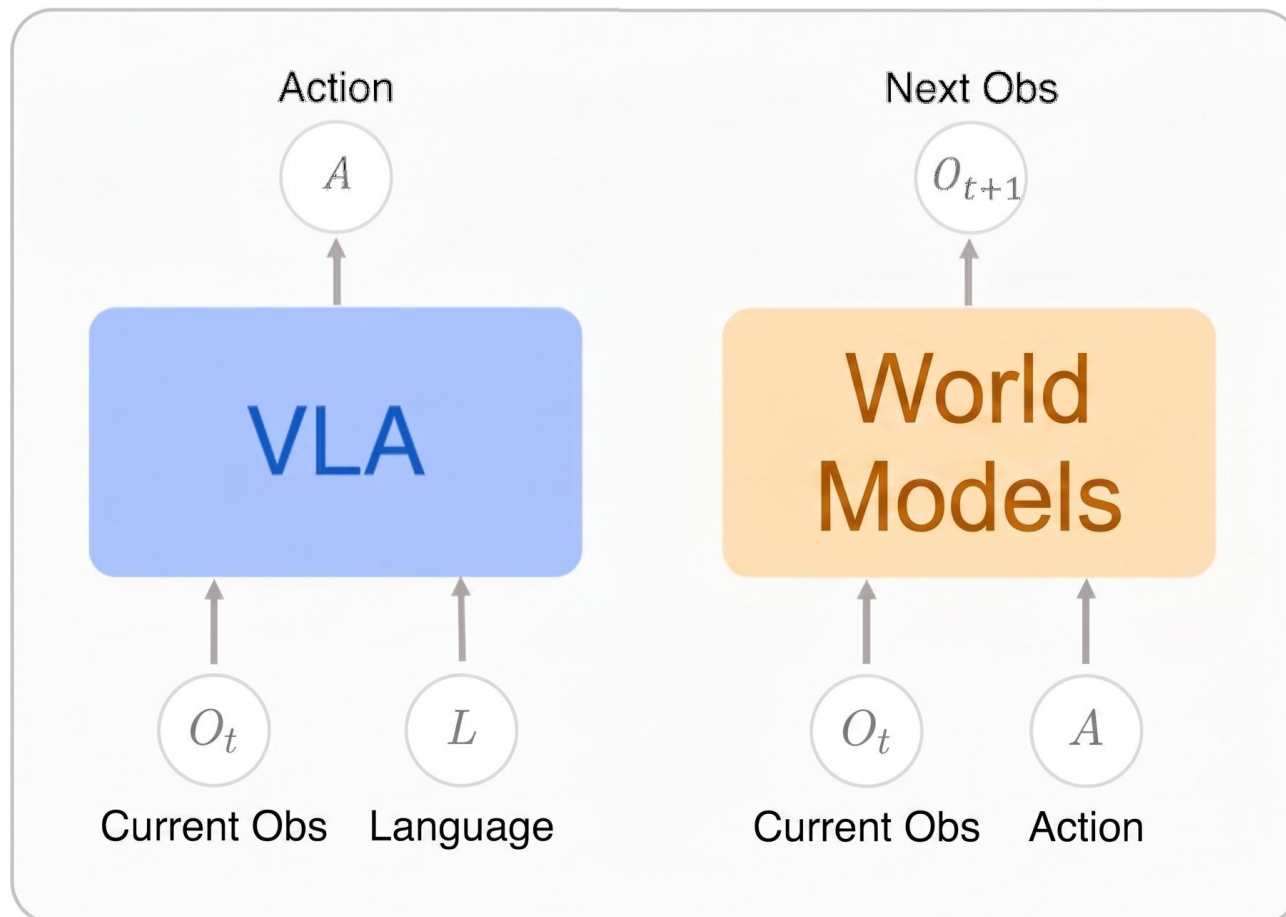
- ✦ Foresight
- ✦ Imagination-driven planning
- ✦ Data amplification

World Models vs. VLA



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□ Presentation Format

□ $\mathcal{L}_{VLA} = \mathbb{E}_{(o,l,a) \sim \mathcal{D}}[-\log p(a|o,l)]$

□ $\mathcal{L}_{WM} = \mathbb{E}_{(o,a,o') \sim \mathcal{D}}[-\log p(o'|o,a)]$

□ Focus

□ “What should I do now?”

□ “If I do X, what happens next?”

AC-WMs & LC-WMs

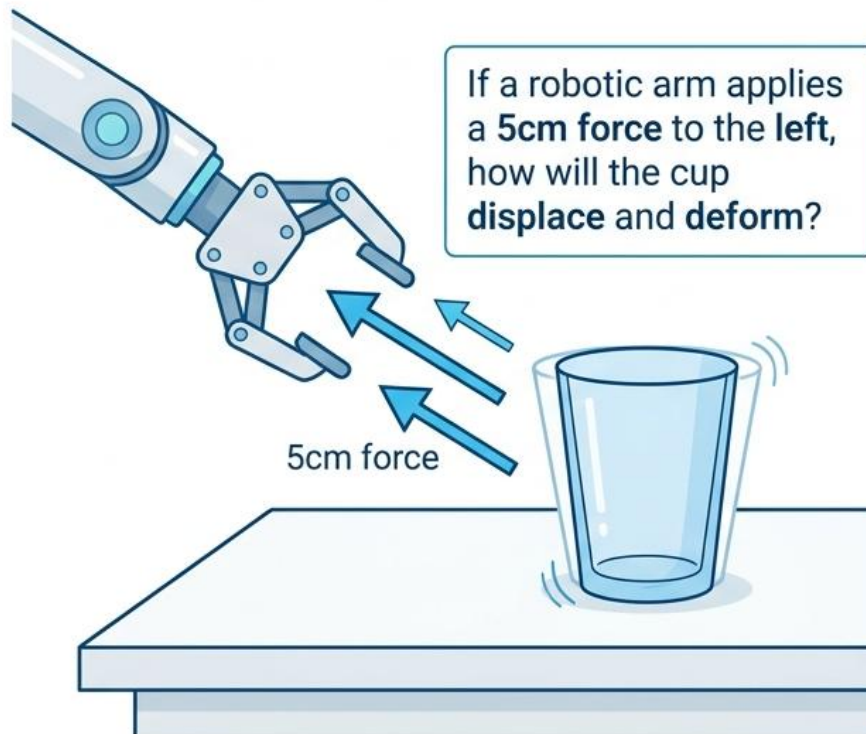


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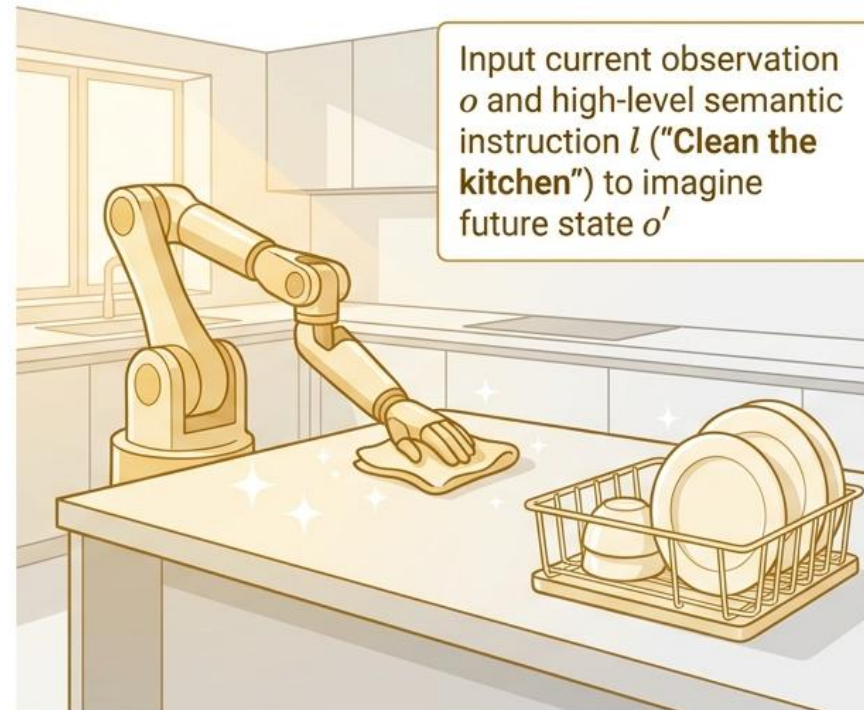
ACTION-CONDITIONED

$$P(o_{t+1} \mid o_t, a_t)$$



LANGUAGE-CONDITIONED

$$P(o_{t+1} \mid o_t, l_t)$$



Why do robots need world models?



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Dimensional Upgrade of Space

- ❑ **LLM:** 1D + t
- ❑ **Autonomous Driving:** 2D + t
- ❑ **Embodied AI Robots:** 3D + t
- ❑ **3D Forward Physics Sandbox:**
World models act as **internal dynamical systems**, accurately simulating 3D space, depth consistency, and rigid-body collisions in their "mind".



Open-World Generalization

- ❑ **OOD Generalization Failure:**
VLA generalization is severely bottlenecked by data.
- ❑ **Bypassing Data Bottlenecks via Video Self-Supervision**
- ❑ **OOD Generalization via Spatiotemporal Priors**



Mitigating Compounding Errors

- ❑ **Reactive Mapping Bottleneck:**
long-horizon reasoning and temporal credit assignment extremely difficult
- ❑ **Compounding Error Avalanche**
- ❑ **World Models Foresight & Correction**



Feature-level Latent Prediction

- ❑ **Computational Waste of Pixel-Level Reconstruction**
- ❑ **Abstract Latent Space:** Predict physical state transitions and causal alignment.

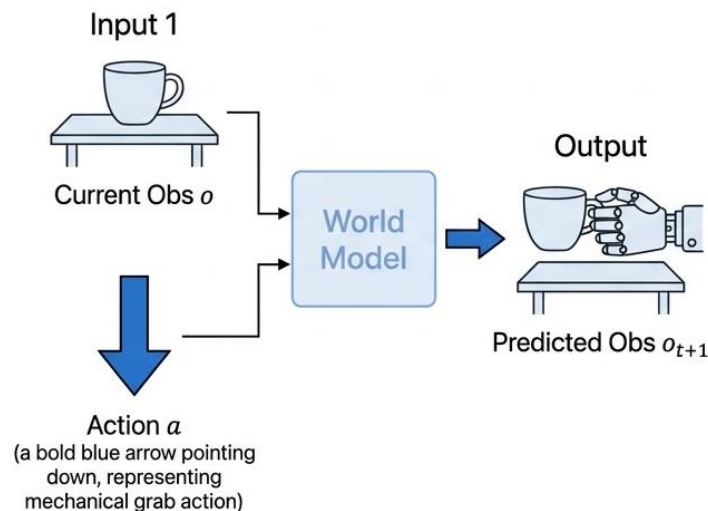
World Models $\neq$ Video Generation Models



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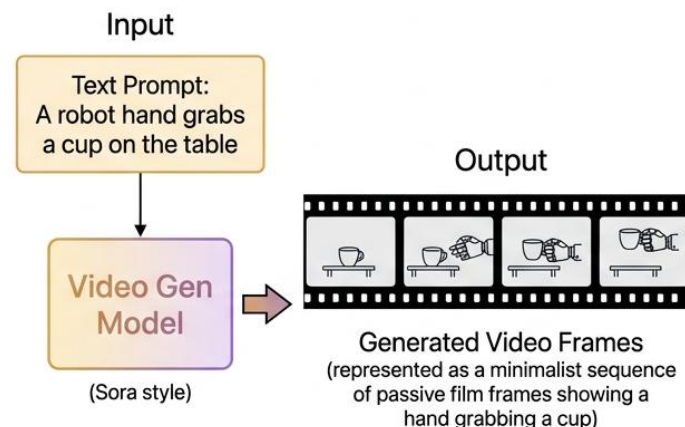
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World Model
(Action-conditioned prediction)



Interactive Foresight for Robot Planning

Video Generation Model
(Passive synthesis)



Passive Text-to-Video Synthesis



World Models

- Action-conditioned Commitment: $P(o'|o, a)$
- Pixels are merely optional proxies, not the objective.
- Explicit & Implicit/Latent-space Prediction



Video Generation Models

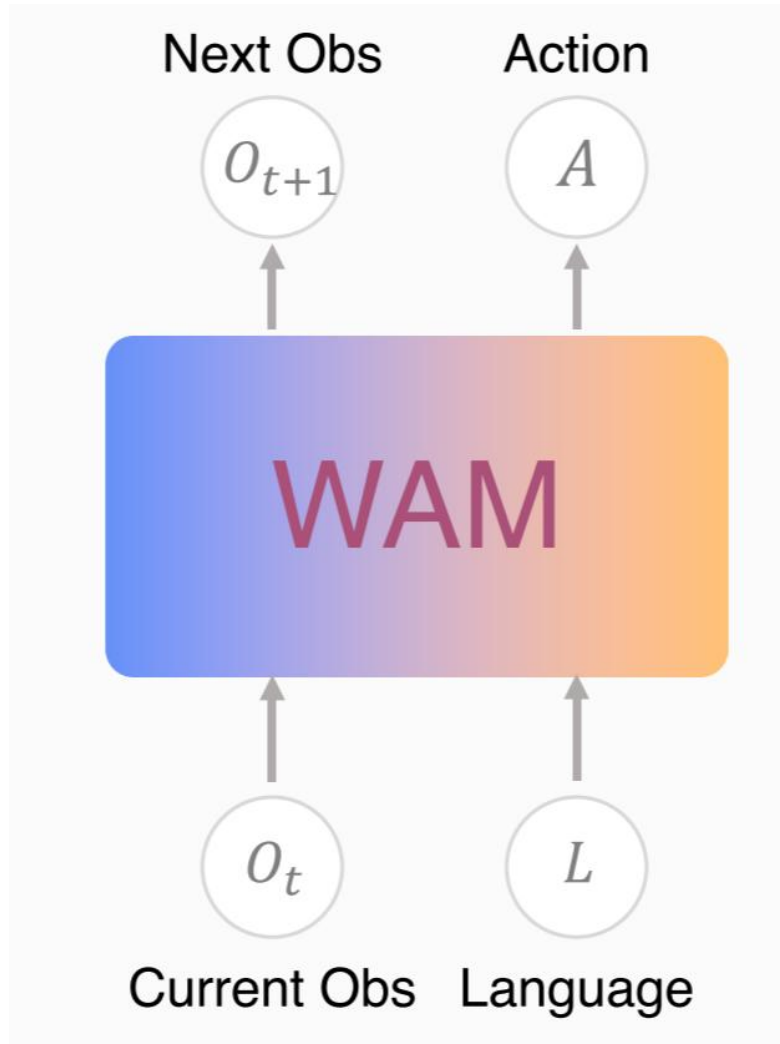
- Passive Visual Continuation: $P(v|o, l)$
- Pixel-level Reconstruction



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World Action Models



- **Presentation Format**

$$\mathcal{L}_{WAM} = \mathbb{E}_{(o,l,o',a) \sim \mathcal{D}} [-\log p(o', a | o, l)]$$

- **Environmental Dynamics Modeling (World Modeling) + Motor Control (Action Generation)**

- **Two Core Criteria**

- **Forward Predictive Modeling**

- Explicit Visual Predictions

- Implicit Physical Representations

- **Coupled Action Generation**

- Achieve deeper **physical understanding** and **stronger zero-shot generalization**.

World Action Models vs. Video Action Models



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Video generation is merely a means of physical inference, rather than the ultimate definition of a world model.

➤ **Superset vs. Subset**

- **WAMs:** A broader, modality-independent superset of predictive agents.
- **VAMs:** Aligning action with synthesized visual futures.

➤ **Multi-modal State vs. Pixel-Bound**

- **WAMs:** Single-image state transitions, Dense point clouds, Tactile, Force feedback
- **VAMs:** Pixel-space video generation

➤ **Causal Dynamics vs. Pixel Synthesis**

- **WAMs:** Underlying Physical Laws & Causal Dynamics
- **VAMs:** Generate visually rich and physically plausible action videos.

02

Two Tech Routes for World Models

How to represent the future state of world models?



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The core divergence lies in state representation: explicit methods generate RGB pixels, while implicit methods use compressed latent features.



Explicit World Models

- ❑ **Definition:** World models directly predict **future state transitions** within the **RGB image** or **video pixel space**.
- ❑ **Core:** Preserve rich spatial geometry, textures, and environmental details.
- ❑ **Limitation:**
 - ❑ Inference Latency Tax
 - ❑ Demand Massive Computational Resources

Explicit Pixel-level Prediction

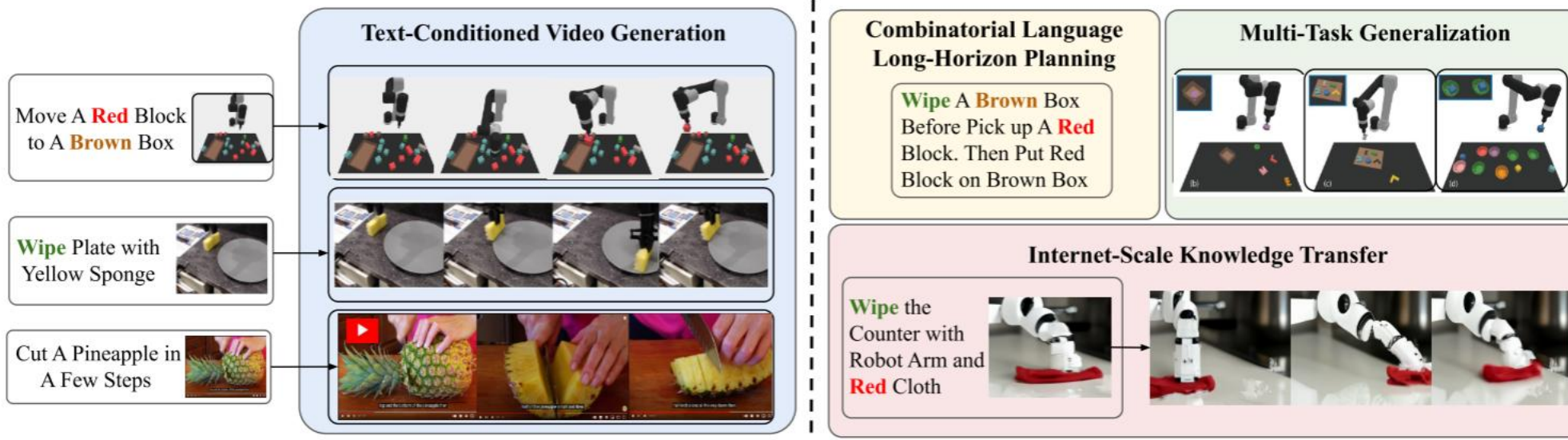


Implicit World Models

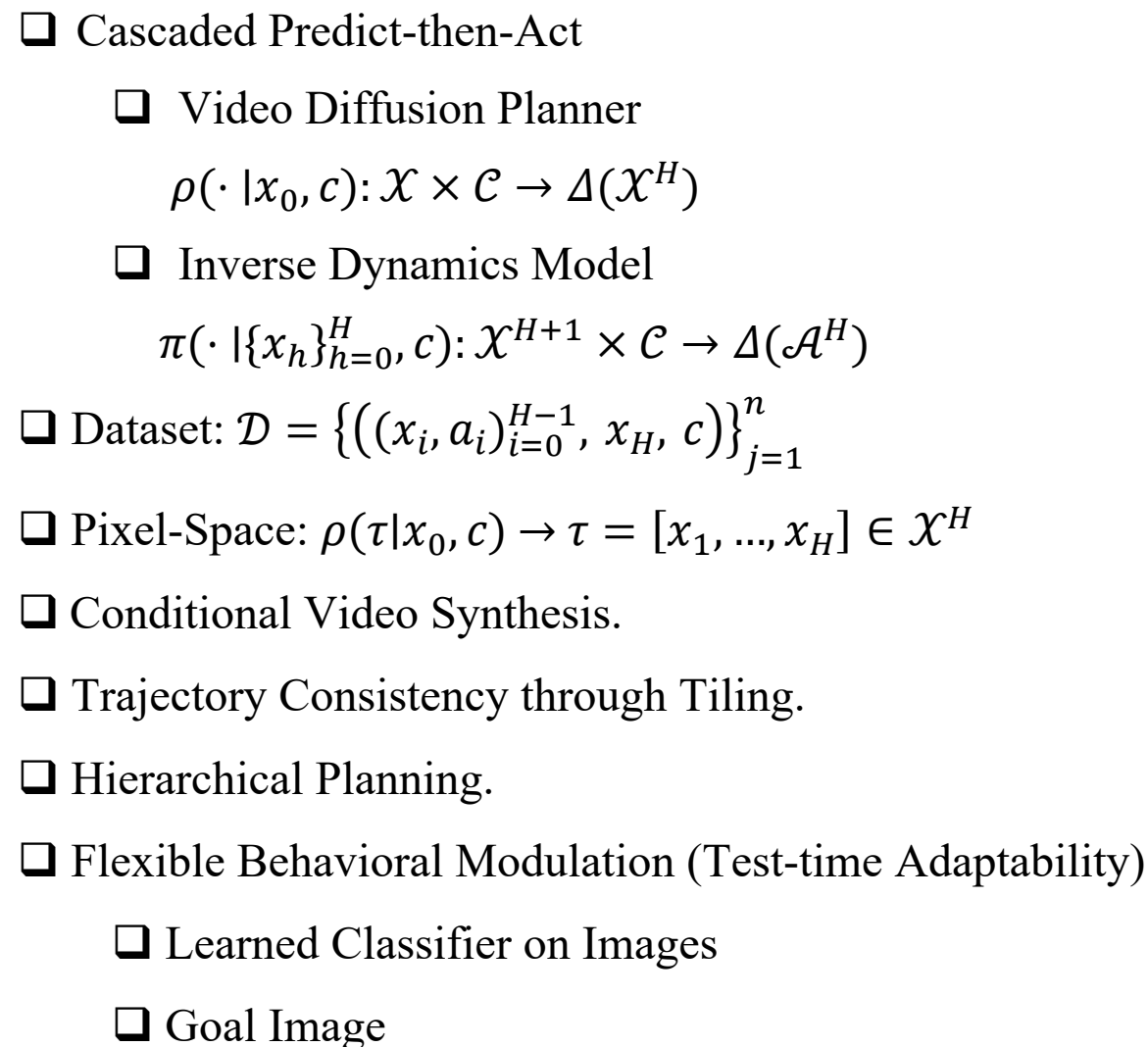
- ❑ **Definition:** Models completely abandon pixel rendering, instead predicting **physical state transitions** directly within an **abstract latent space**.
- ❑ **Core:** Predict the evolution of high-dimensional physical features via latent alignment.
- ❑ **Limitation:**
 - ❑ Lack of Interpretability
 - ❑ Potential Loss of Fine-grained Details

Implicit Latent-space Dynamics Models

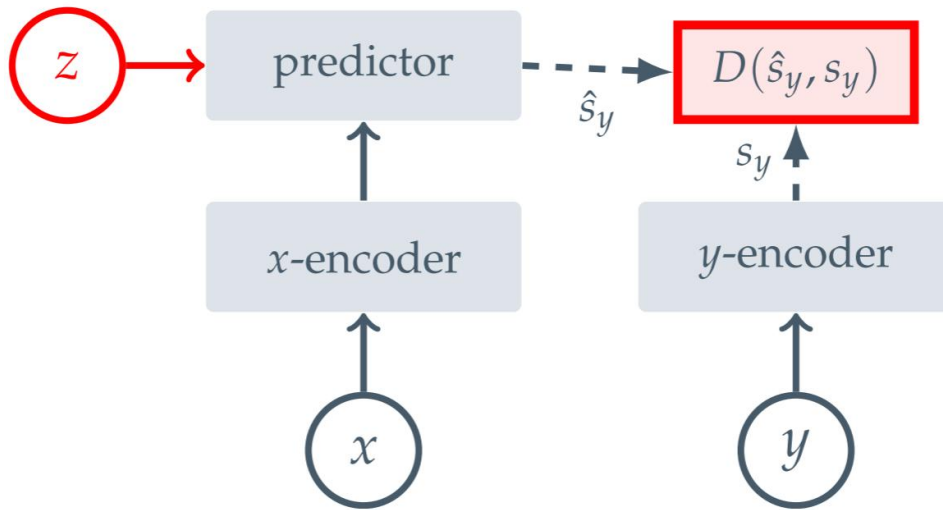
Explicit Route: UniPi Case Study



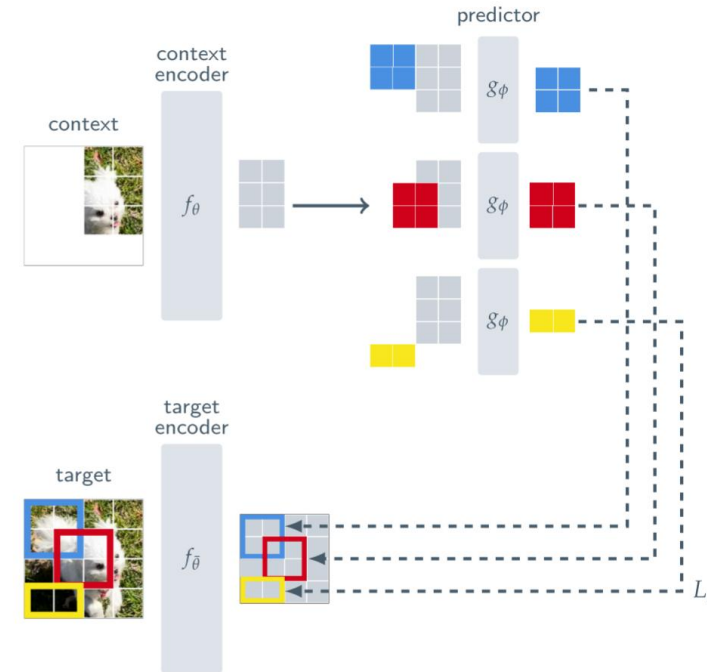
- ❑ **Universal Policy:** Sequential Decision Making Problem → Text-Conditioned Video Generation Problem
- ❑ **Sources of Data:** Simulated, Real Robots, YouTube
- ❑ **Advantages:**
 - ❑ Combinatorial Generalization
 - ❑ Action Planning (Video Generation Procedure)
 - ❑ Multi-task Learning
 - ❑ Internet-Scale Knowledge Transfer



Implicit Route: Predictive Representation Learning via I-JEPA



(a) Joint-Embedding Predictive Architecture



(b) Image-based Joint-Embedding Predictive Architecture

❑ **Asymmetric Architecture:** Context Encoder (f_θ) & Target Encoder ($f_{\bar{\theta}}$) & Predictor (g_ϕ) (ViT)

❑ **Targets:** Image $y \rightarrow$ Sequence of N Non-overlapping Patches $\rightarrow f_{\bar{\theta}} \rightarrow \mathbf{s}_y = \{\mathbf{s}_{y_1}, \dots, \mathbf{s}_{y_N}\} \rightarrow$ Randomly sample M blocks $\mathbf{s}_y(i) = \{\mathbf{s}_{y_j}\}_{j \in B_i}$ ($M = 4$) B_i : Mask corresponding of the i^{th} block

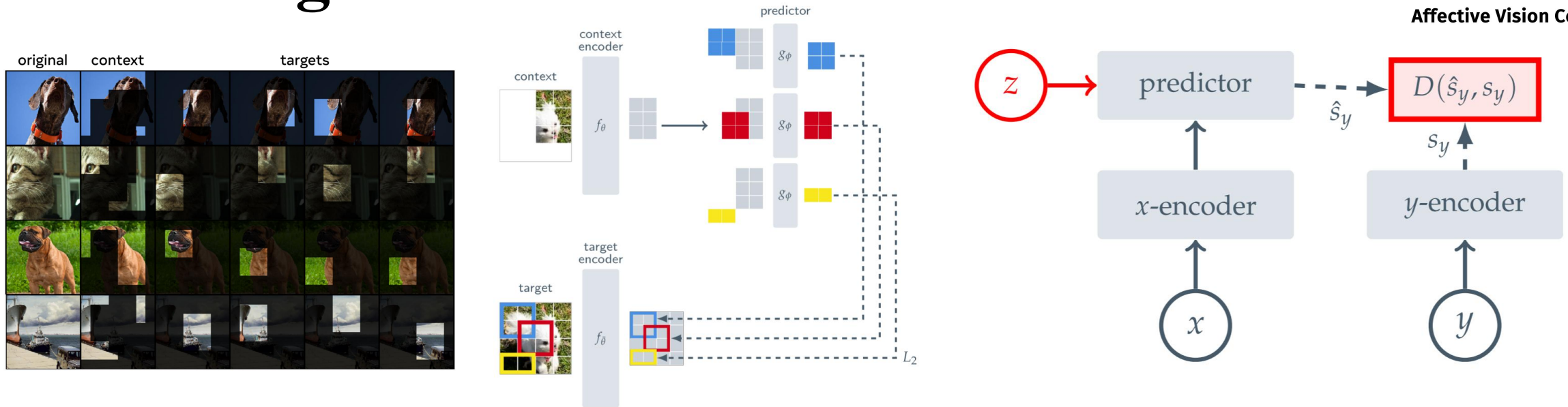
❑ **Context:** Sample a Single Block x (1:1) $\rightarrow$ Remove any overlapping regions from the context block $\rightarrow f_\theta \rightarrow$ Patch-level representation $\mathbf{s}_x = \{\mathbf{s}_{x_j}\}_{j \in B_x}$

Implicit Route: Predictive Representation Learning via I-JEPA



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□ **Prediction:** $s_x \rightarrow g_\phi(\cdot, \cdot) \rightarrow \hat{s}_y(i) = \{\hat{s}_{y_j}\}_{j \in B_i} = g_\phi(s_x, \{m_j\}_{j \in B_i}) \rightarrow \hat{s}_y(1), \dots, \hat{s}_y(M)$

□ **L₂ Loss:** $\frac{1}{M} \sum_{i=1}^M D(\hat{s}_y(i), s_y(i)) = \frac{1}{M} \sum_{i=1}^M \sum_{j \in B_i} \|\hat{s}_{y_j} - s_{y_j}\|_2^2$ **(Latent Space)**

□ **Gradient-based optimization:** θ & ϕ

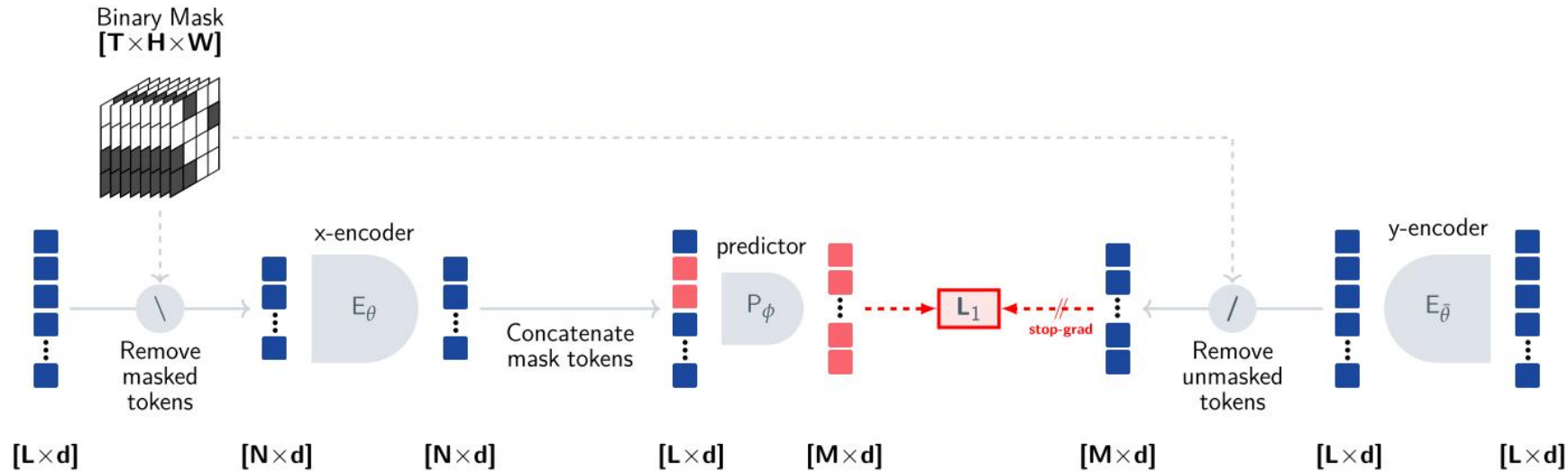
□ **Exponential Moving Average:** Target Encoder $\bar{\theta}_t = m_t \bar{\theta}_{t-1} + (1 - m_t) \theta_t$ $m_t: 0.996 \rightarrow 1.0$

Implicit Route: Predictive Representation Learning via V-JEPA



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□ $T \times H \times W \rightarrow \text{Patch: } 2 \times 16 \times 16 \rightarrow L \text{ Spatio-temporal Tokens} \rightarrow L \times d$

□ Sample y from a video: Sample several (possibly overlapping) spatially continuous blocks with various aspect ratios and repeat the spatial blocks **across the entire temporal dimension of the video**.

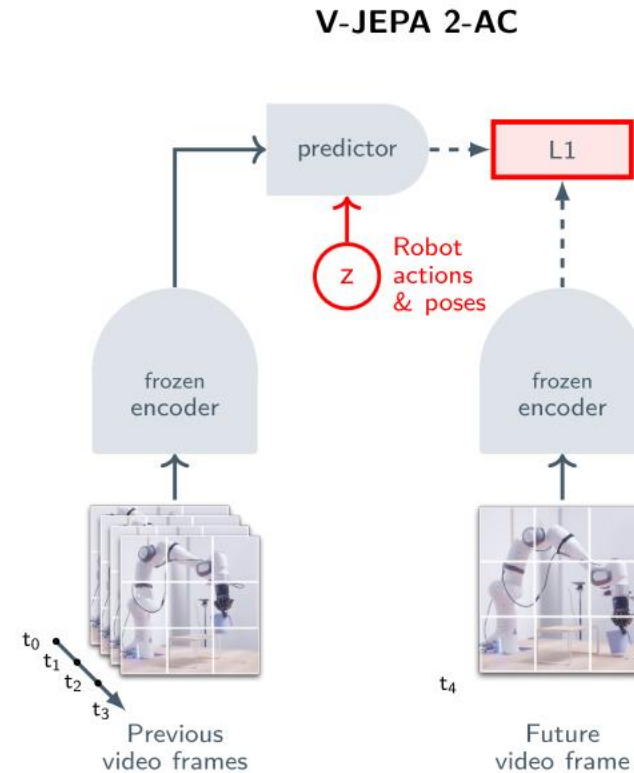
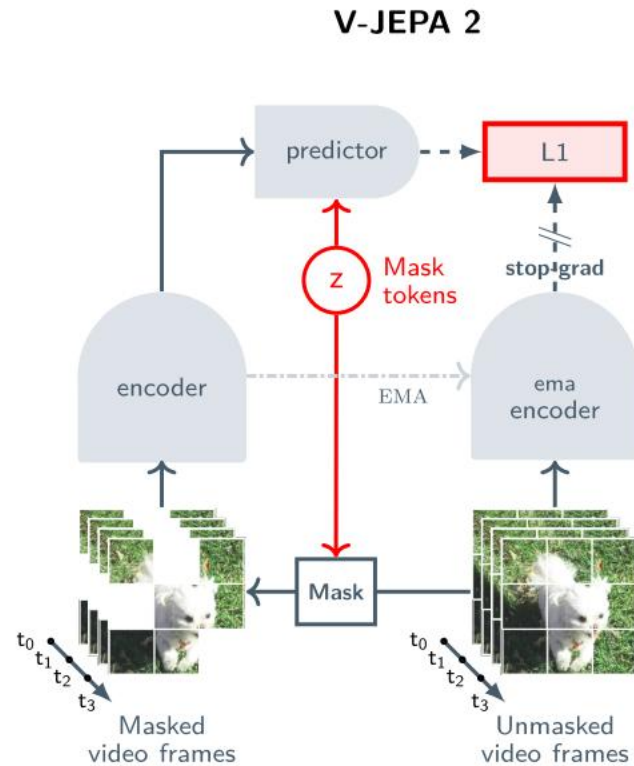
□ **L₁ Loss:** $\min_{\theta, \phi} \|P_\phi(E_\theta(x), \Delta_y) - \text{sg}(\tilde{E}_\theta(y))\|_1$ **(Latent Space)**

V-JEPA 2-AC



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❑ Freeze The Video Encoder

❑ Learn a New Action-conditioned Predictor

❑ Autoregressive Feature Prediction objective: Conditioned on Past Video Frames & Actions & End-effector States → Predict the **Representations of Future Video Frames**

03

Architecture of World Action Models

Two Primary Paradigms: Cascaded WAM vs. Joint WAM



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*These World Action Model architectures can be categorized into two primary paradigms based on their **structural flow** and **corresponding training regimes**.*

Cascaded

Think Before You Act

- 01 **Mathematical Essence:** $\mathcal{L}_{Cascaded} = p(o', a|o, l) = p(a|o', o, l) p(o'|o, l)$
- 02 **Separated Training Process:** The World model and the action decoder are optimized as distinct modules.
- 03 **Representative Models:** **UniPi**, VPP, $\pi_{0.7}$, etc.

Joint

Think While You Act

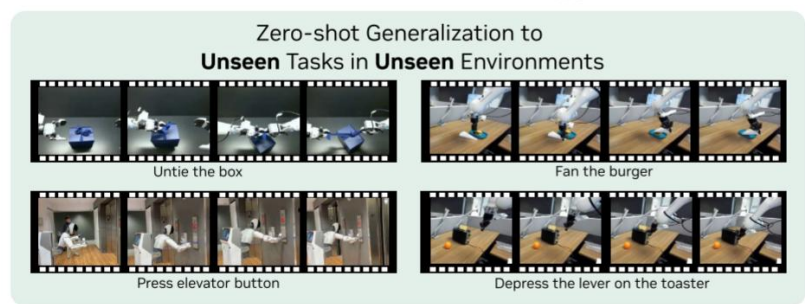
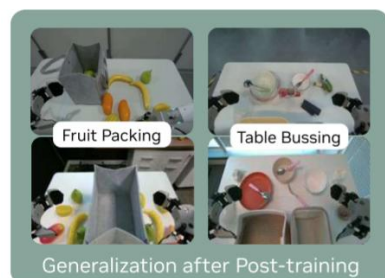
- 01 **Mathematical Essence:** $\mathcal{L}_{Joint} = p(o', a|o, l)$
- 02 **Joint Training Process:** World modeling and action generation undergo joint training under a unified objective, forcing the model to internalize the **causal interdependencies** between **environmental dynamics** and **control signals**.
- 03 **Representative Models:** **DreamZero**, UWM, WorldVLA, etc.

Joint WAM: DreamZero



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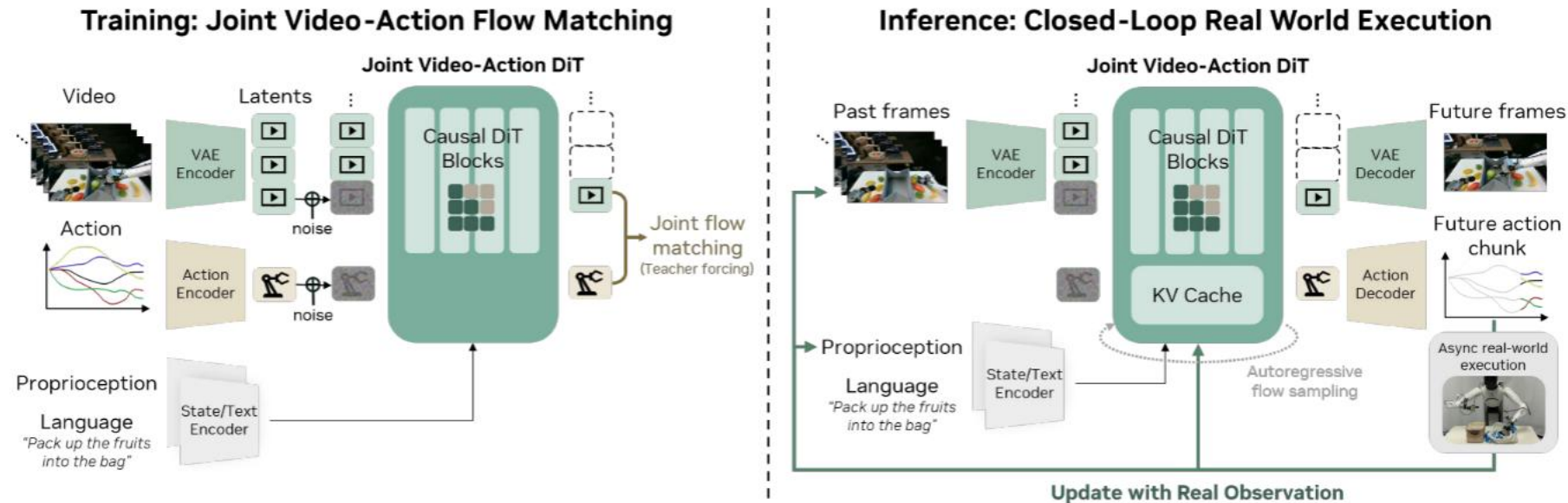
- ❑ Effective learning from diverse, non-repetitive data (**Task Diversity > Repetition**)
- ❑ Open-world generalization (**Zero-shot generalization to new tasks in new environments**)
- ❑ Cross-embodiment learning from video-only data (**humans (12 minutes) or other robots (20 minutes)**)
- ❑ Few-shot adaptation to new robots (**Adapt to an entirely new robot with only 30 minutes of play data**)

Joint WAM: DreamZero



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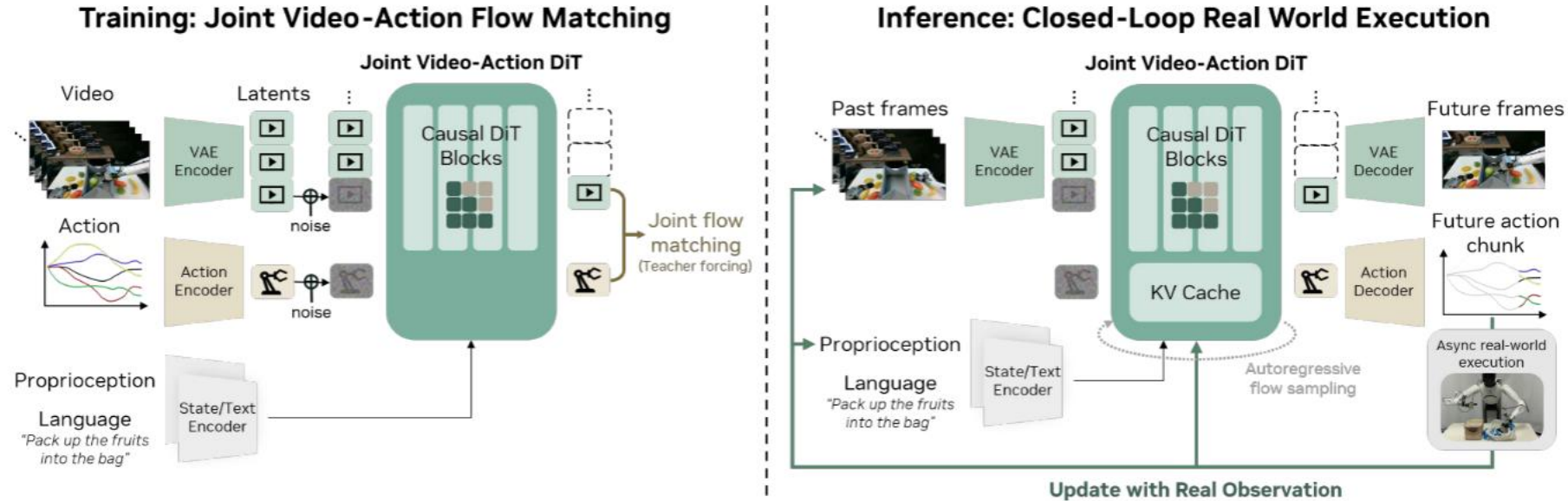
- ❑ Architecture: Autoregressive Diffusion Transformer
- ❑ Multi-view Concatenation: Concatenate all views into a single frame
- ❑ Video → Frozen VAE Encoder → Visual Context
- ❑ Language Instructions → Text Encoder → Text Embedding
- ❑ Proprioception → State Encoder → State Embedding

Joint WAM: DreamZero



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□ Autoregressive Modeling Only for the Video Modality: $\pi_0(o_{l:l+H}, a_{l:l+H} | o_{0:l}, c, q_l) = \underbrace{\pi_0(o_{l:l+H} | o_{0:l}, c, q_l)}_{\text{video prediction}} \underbrace{\pi_0(a_{l:l+H} | o_{0:l+H}, q_l)}_{\text{IDM}} \text{DreamZero}$

□ Teacher-forcing: Model is trained to denoise the noisy current chunk conditioned on the clean previous chunks $C_k = \{(z_1^j, a_1^j)\}_{j=1}^{k-1}$, where z_1^k and a_1^k are a clean video latent vector and a normalized action.

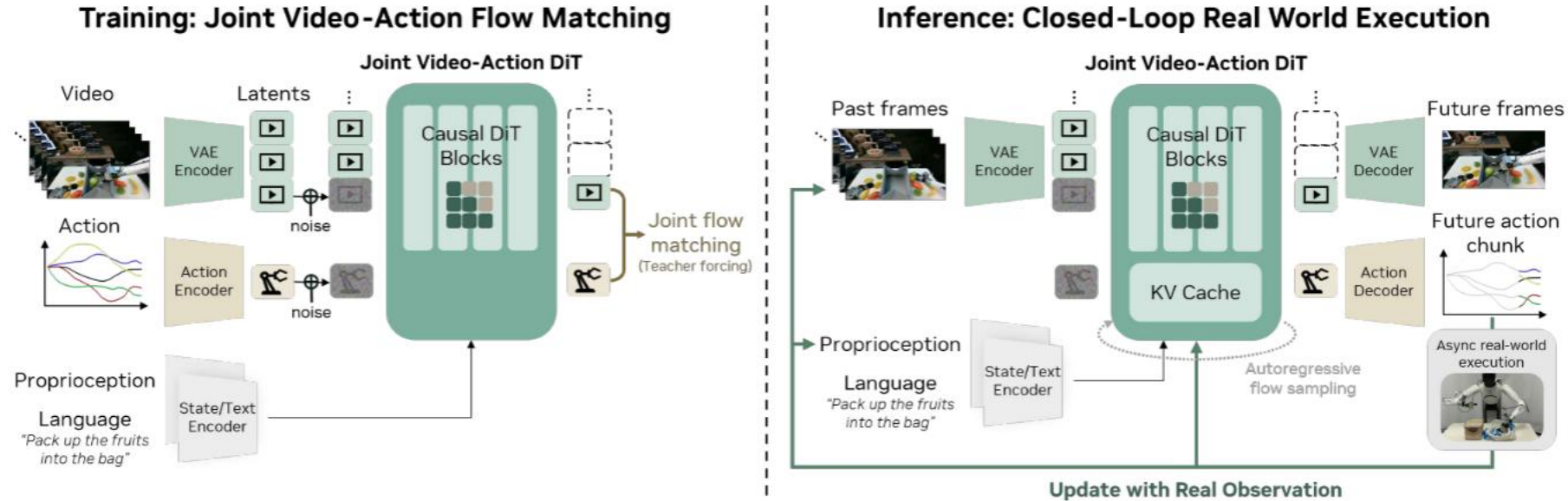
□ Inference: Enables Faster Inference Speed by Utilizing KV-cache

Joint WAM: DreamZero



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□ Linear Interpolations between Clean Vectors and Random Gaussian Noises: $z_{t_k}^k = t_k z_1^k + (1 - t_k) z_0^k$, $a_{t_k}^k = t_k a_1^k + (1 - t_k) a_0^k$, $z_0^k \sim \mathcal{N}(0, I)$, $a_0^k \sim \mathcal{N}(0, I)$

□ Flow Matching Loss:

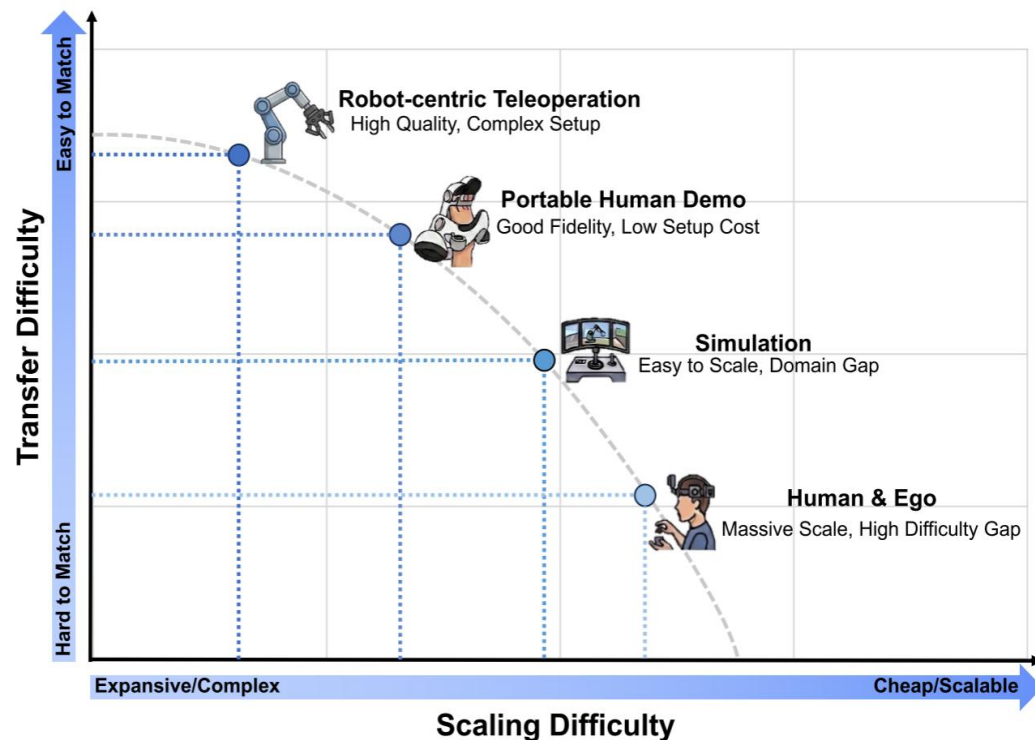
$$\mathcal{L}(\theta) = \mathbb{E}_{z, a, \{t_k\}} \left[\frac{1}{K} \sum_{k=1}^K w(t_k) \|u_{\theta}([z_{t_k}^k, a_{t_k}^k]; \mathcal{C}_k, c, q_k, t_k) - v^k\|^2 \right], \mathcal{C}_k = \left\{ (z_1^j, a_1^j) \right\}_{j=1}^{k-1},$$

$$v^k := [z_1^k, a_1^k] - [z_0^k, a_0^k]$$

04

Data & Evaluation

Embodied Data for training WAMs



❑ High-Fidelity Robot-Centric Teleoperation

- ✓ Core Role: Provides precise kinematic grounding.

❑ Agile Portable Human Demonstrations (UMI-style)

- ✓ Core Role: Acts as a critical bridge connecting human dexterity and real-world interactions.

❑ Highly scalable Simulation Data

- ✓ Core Role: Break the bottlenecks of real-robot data collection.

❑ Broad-Coverage Human- and Ego-Centric Data

- ✓ Core Role: Internalizes unbounded passive world dynamics knowledge via passive visual observations.

Datasets Examples

❑ DROID

DROID Dataset Visualizer (Showing random subset of 1k trajectories from DROID)

Scene Types

- ☒ All (1000)
- ☐ Industrial office (586)
- ☐ Kitchen (252)
- ☐ Office (82)
- ☐ Living room (27)
- ☐ Dining room (17)
- ☐ Bathroom (14)
- ☐ Bedroom (11)
- ☐ Hallway (8)
- ☐ Laundry room (3)
- ☐ Other (0)

Object Types

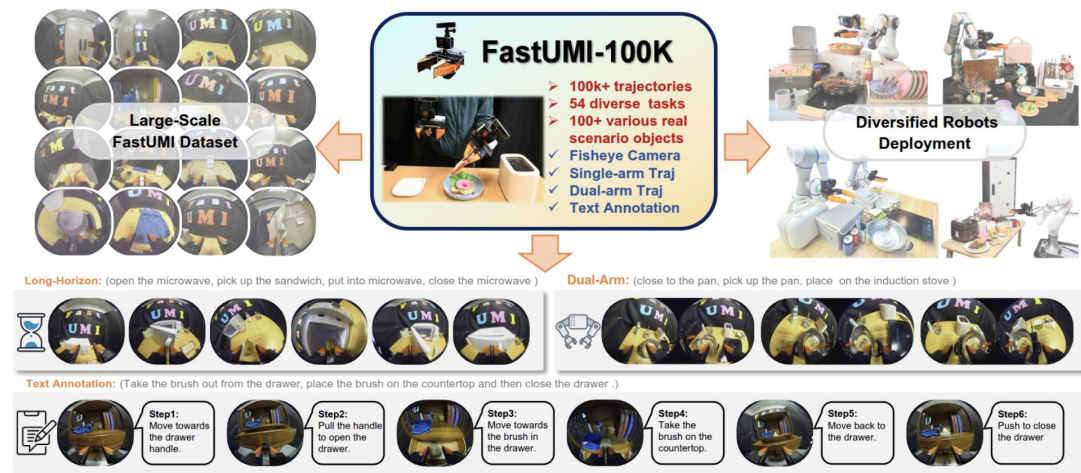
- ☒ All (1000)
- ☐ Marker (163)
- ☐ Cloth (72)
- ☐ Cup (36)
- ☐ Object (32)
- ☐ Bottle (31)
- ☐ Block (30)
- ☐ Lid (24)
- ☐ Drawer (23)
- ☐ Bowl (20)
- ☐ Other (569)

Task Types

- ☒ All (1000)
- ☐ Put (237)
- ☐ Move (141)
- ☐ Pick (133)
- ☐ Remove (95)
- ☐ Take (57)
- ☐ Close (55)
- ☐ Open (54)
- ☐ Place (28)
- ☐ Turn (26)
- ☐ Other (174)

Showing 40 matches randomly sampled from 1000 total matches

❑ FastUMI-100K



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❑ RoboTwin 2.0

50 Dual-arm Tasks

731 Diverse Objects

5 Embodiments

Rich Diversity

Benchmark

Large Dataset Scale

RoboTwin

Performance

Sim2Real

4.7x ↑ 30% 42%

9% 47.4%

Auto Code Generation

71.3% (ours)

60.4% 63.9%

● RoboTwin 1.0 Vanilla
● RoboTwin 1.0+FB
● RoboTwin 1.0+FB+MLLM
● RoboTwin 2.0

Domain Randomization

Randomization

Unseen Real-World

Open Source

RoboTwin Object Dataset

Community LeaderBoard

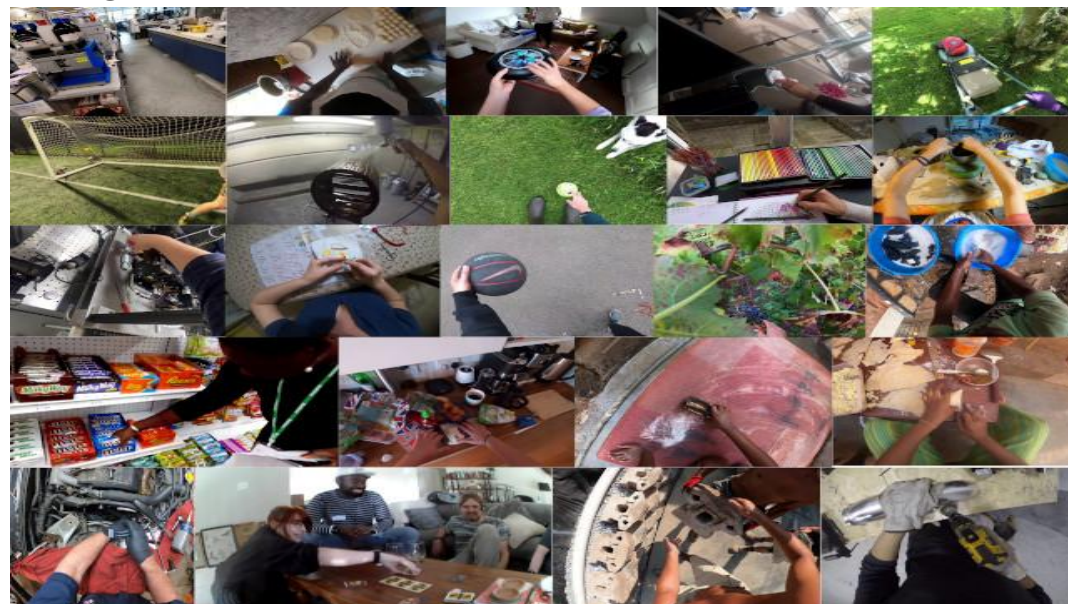
Deploy Diverse Policies

Pre-Collected 100k+ Traj.

Friendly Codes & Tutorial

Generate Any Dual-Arm Robot Manip Data

❑ Ego4D



How to Validate Real-World Effectiveness of WAMs?



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❑ World Modeling Capability

❑ Visual Fidelity:

- Low-Level Fidelity: PSNR, SSIM
- Perceptual and Semantic Consistency: LPIPS, DreamSim, DINO Similarity
- Distribution-Level Realism and Temporal Quality: FVD

❑ Physical Commonsense: WorldModelBench, WorldScore, etc.

❑ Action Plausibility: WorldSimBench, etc.

❑ Action Policy Capability

❑ Task Success Rate

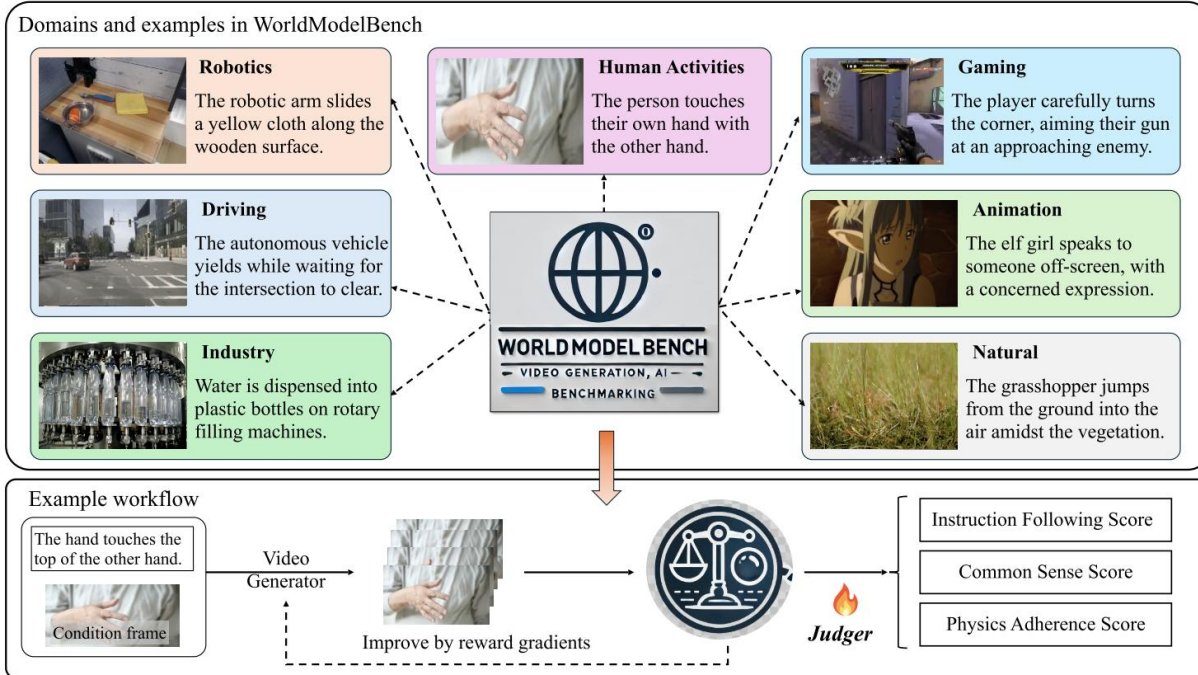
❑ Average Task Progress

Physical Commonsense: WorldModelBench



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A robotic arm hovers above a wooden table, ready to interact with the towel.



Model A



Model B



VBench

	Smoothness	Consistency	Image Quality	Win Rate
Model A	✓	✓	✓	0.5
Model B	✓	✓	✓	0.5



Ours

	...	Newton's Law	Gravity	Win Rate
Model A	...	✓	✗	0.2
Model B	...	✓	✓	0.8

❑ **Details:** 350 image-text condition pairs, 7 application-driven domains, 56 diverse subdomains, supports T2V & I2V, 67K human labels

❑ **Assessment Principles:** Instruction Following (Scores 0-3) & Physics Adherence (Total Scores 0-5) & Commonsense (Frame-Wise Quality & Temporal Quality) (total scores 0-2)

❑ **A Reliable World Modeling Judger (Reward Model)**

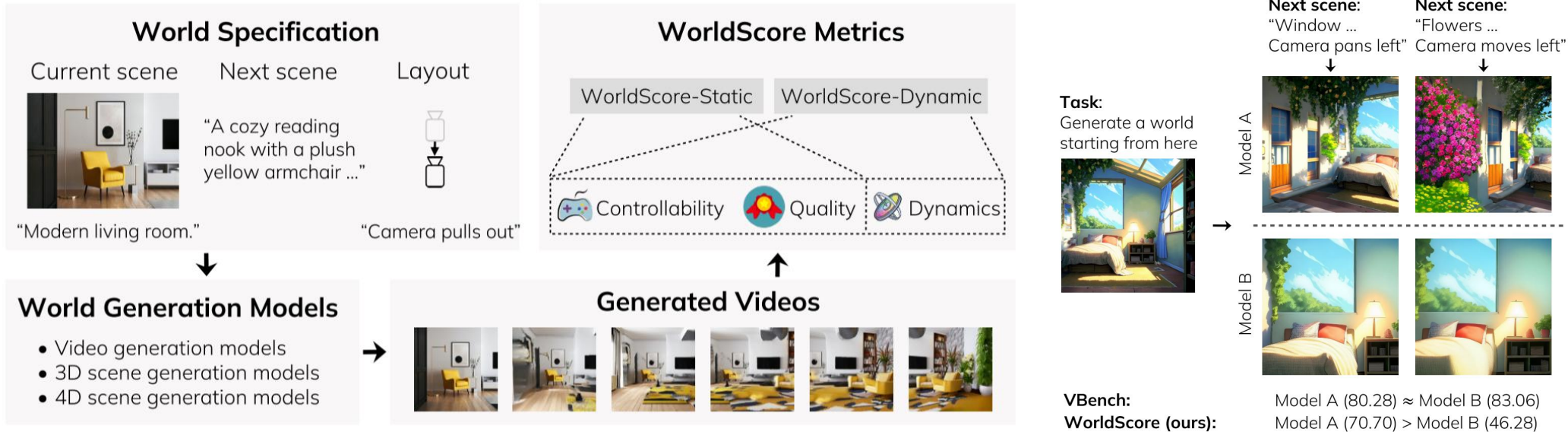
$$J(\theta) = \mathbb{E}_{c \sim D_c, x_0 \sim p_\theta(x_0|c)} \left[\sum_{g \sim G} R(x_0, c, g) \right]$$

Physical Commonsense: WorldScore



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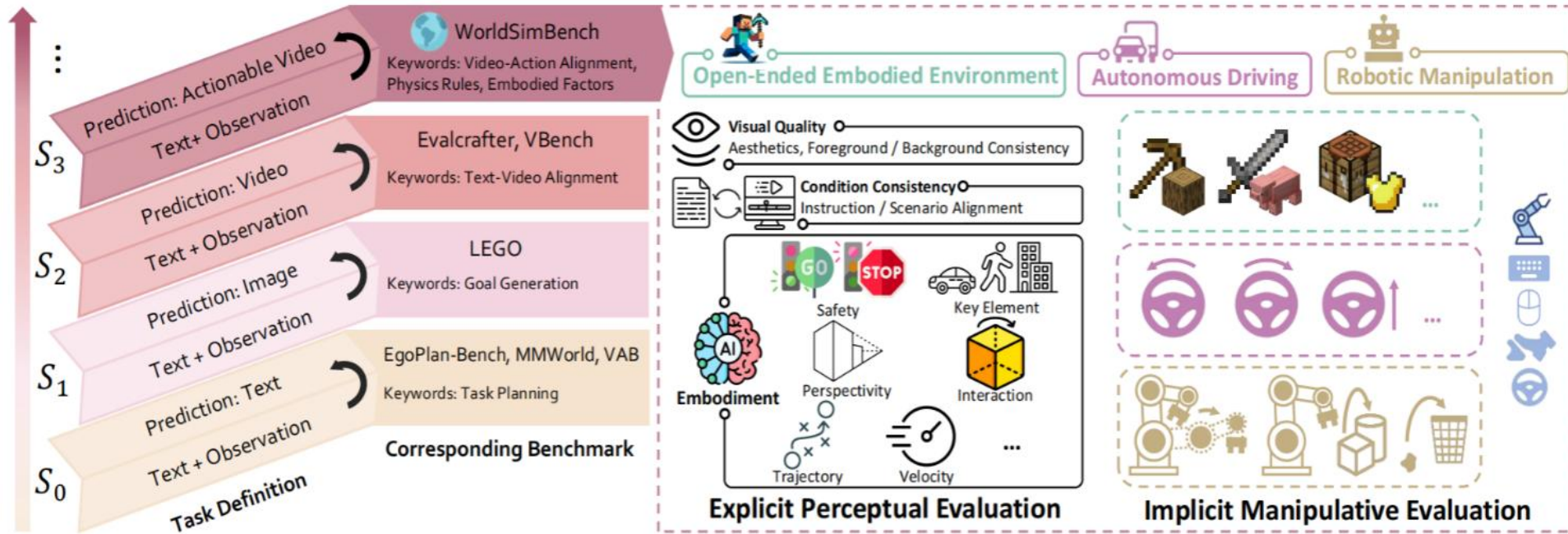
- ❑ **Motivation:** Existing benchmarks lack explicit spatial-layout control, camera specifications and reference images, which limits evaluations to single scenes.
- ❑ **World Generation** → A Sequence of Next-Scene Generation Tasks. (current scene, next scene, layout)
- ❑ **Evaluation Scope:** 3D, 4D, T2V, I2V
- ❑ **10 Metrics:** **Controllability** (Camera Controllability, Object Controllability, Content Alignment), **Quality** (3D Consistency, Photometric Consistency, Style consistency, Subjective quality), **Dynamics** (Motion accuracy, Motion magnitude, Motion smoothness)
- ❑ **By Application Scenarios:** WorldScore-Static & WorldScore-Dynamic

Action Plausibility: WorldSimBench



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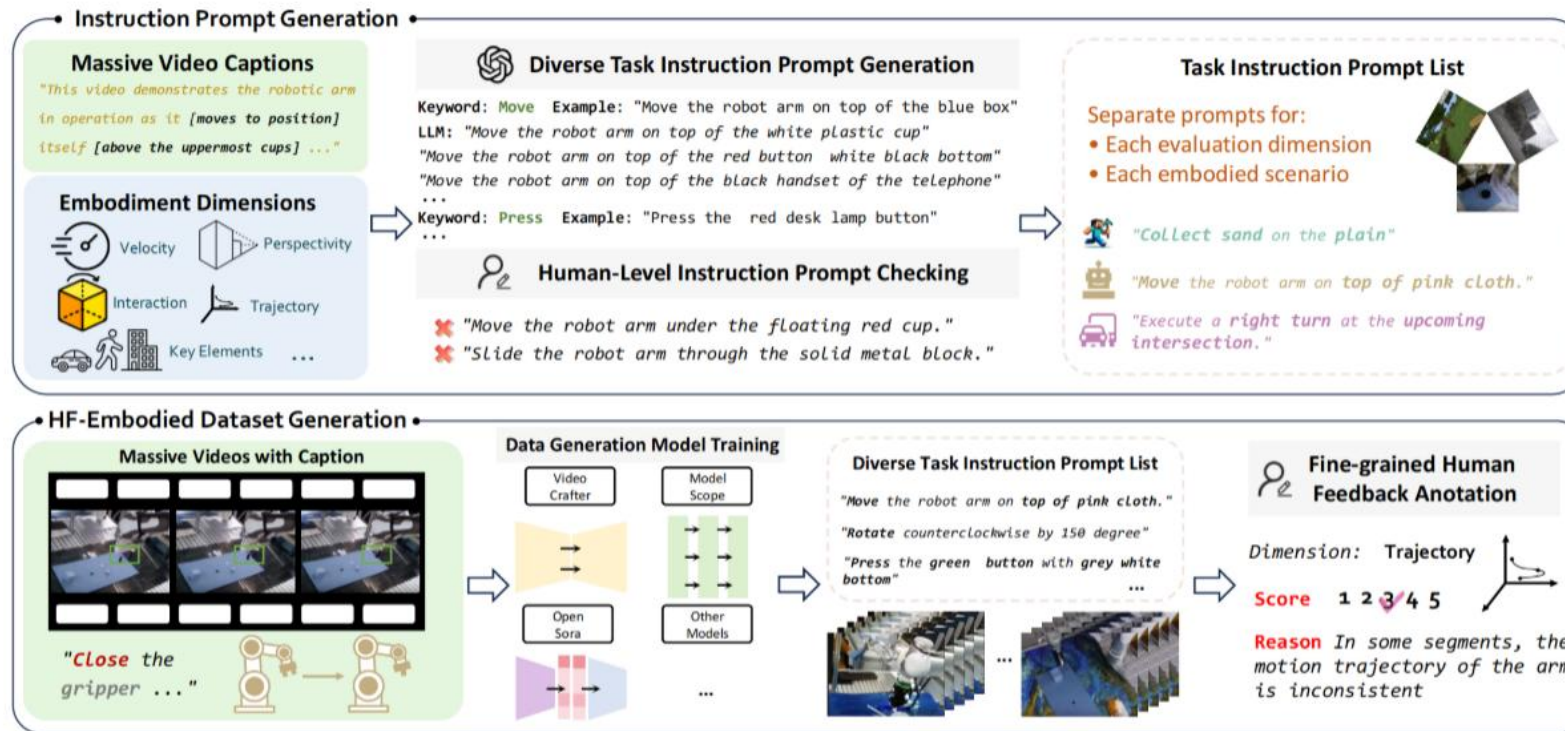
- ❑ **Hierarchical Capabilities of the Predictive Models:** $S_0 \rightarrow S_3$
- ❑ **Dual Evaluation Framework:** Explicit Perceptual & Implicit Manipulative
- ❑ **Three Embodied Scenarios:** Open-Ended Embodied Environment, Autonomous Driving, Robot Manipulation
- ❑ **Focus:** Video Generation & Action Transformation

Action Plausibility: WorldSimBench



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✓ **Explicit Perceptual Evaluation**

❑ **Instruction Prompt Generation**

❑ **HF-Embodied Dataset Generation**

❑ **Video Assessment Dataset:** HF-Embodied Dataset (35,701 entries)

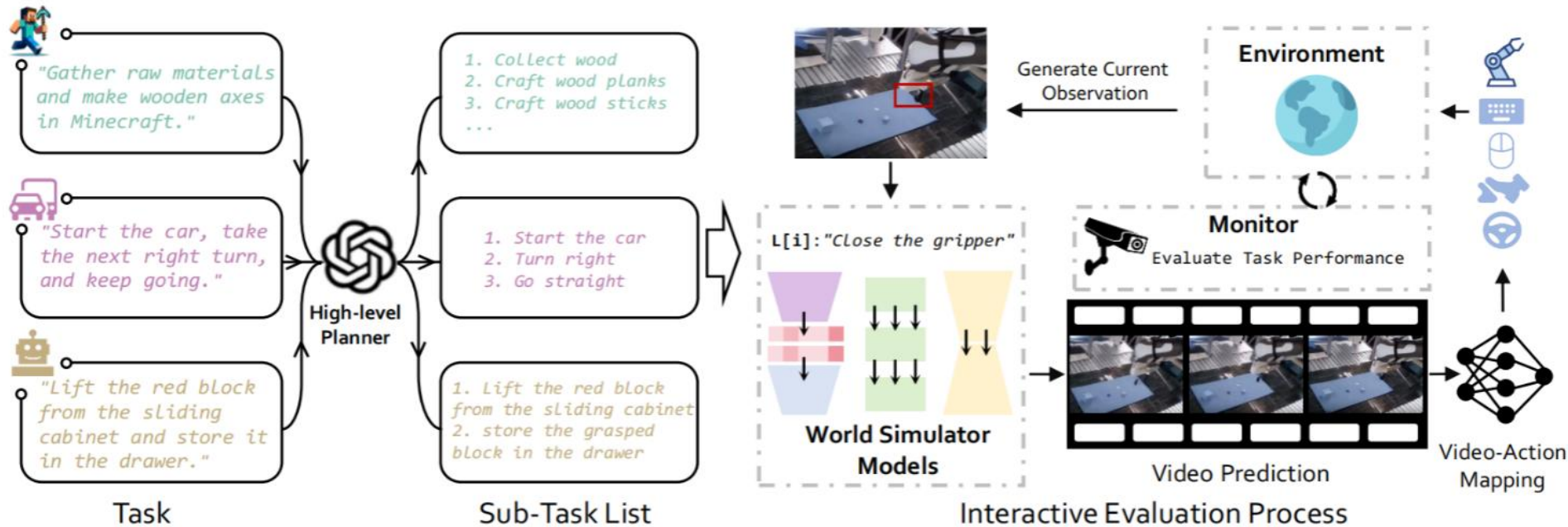
❑ **Train a Human Preference Evaluator:** Aligns with human perception and explicitly assesses the visual fidelity of World Simulators.

Action Plausibility: WorldSimBench



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✓ **Implicit Manipulative Evaluation**

- ❑ Assess the Video-Action Consistency of World Simulators
- ❑ Embodied Tasks in Different Scenarios → Executable Sub-tasks
- ❑ Current Instructions + Observations → Predicted Videos
- ❑ Video → Pre-trained IDM or a Goal-based Policy → Action
- ❑ Evaluation: Success Rates of Various Embodied Tasks



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References

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THANKS